MIL-C-83360C
27 March 1975

SUPERSEDING
MIL-C-83360B(USAF)
23 October 1973

MILITARY SPECIFICATION

CLEANING AND LUBRICATING COMPOUND, ELECTRICAL CONTACT

This specification is approved for use by all Departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the two types of cleaner-lubricating compounds and one type of cleaner-non-lubricant compound designed for use in cleaning and lubricating electrical contacts.

1.2 Classification. The compounds shall be of the following types, as specified:

Type I - Cleaner-lubricant compounds which have no deleterious effect on potting compounds, rubber materials, and insulations.

Type II - Cleaner-lubricant compounds which may have deleterious effect, to some degree, on certain potting compounds, rubber materials and insulating materials.

Type III - Cleaning compounds which have no deleterious effect on potting compounds, rubber materials, and insulations.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards of the issues in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

O-T-620 Trichloroethane 1, 1, 1, Technical, Inhibited (Methyl Chloroform)

FSC 6850

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BB-F-1421	Fluorocarbon Refrigerants
QQ-A-250/4	Aluminum Alloy 2024, Plate and Sheet
QQ-A-671	Anode, Cadmium
QQ-B-626	Brass, Leaded and Nonleaded, Rod, Shaped Forgings, and Flat Products With Finished Edges (Bar and Strip)
QQ-C-576	Copper Flat Products With Slit, Slit and Edge-rolled, Sheared, Sawed or Machined Edges, (Plate, Bar Sheet and Strip)
QQ-M-44	Magnesium Alloy Plate and Sheet (AZ31B)
QQ-Z-285	Zinc, Anodes
TT-T-291	Thinner, Paint, Volatile Spirits (Petroleum-spirits)
VV-D-1078	Damping Fluid, Silicone Base (Dimethyl Polysiloxane)
PPP-B-636	Box, Fiberboard
PPP-C-96	Can, Metal, 28 Gage and Lighter

Military

MIL-C-81302	Cleaning Compound, Solvent, Trichlorotrifluoroethane

STANDARDS

Federal

FED-STD-102	Preservation, Packaging and Packing Levels
FED-STD-123	Marking for Domestic Shipment (Civilian Agencies)

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The material shall consist of a mixture of cleaning solvent, contact lubricant and propellant, contained in a pressurized container.

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3.2 Composition. The cleaner-lubricant or cleaner material composition shall be as specified.

3.2.1 Type I

Trichlorotrifluoroethane, Type II	60 to 65 percent by weight of total filling mixture.
Silicone fluid	3 to 5 percent by volume of usable product.
Propellant	35 to 40 percent by weight of total mixture.

3.2.2 Type II

Tricholorethane 1, 1, 1	65 to 70 percent by weight of total filling mixture.
Silicone fluid	3 to 5 percent by volume of usable product.
Propellant	30 to 35 percent by weight of total mixture.

3.2.3 Type III

Trichlorotrifluorethane Type II	70 to 75 percent by weight of total filling mixture
Propellant	25 to 30 percent by weight of total wixture.

3.3 Chemical and physical requirements

3.3.1 Lubricant. Lubricant shall conform to VV-D-1078 and the viscosity shall be from 100 to 500 centistokes at 25°C (77°F).

3.3.2 Cleaning solvent.

Type I	-	Solvent shall conform to 3.2 of MIL-C-81302, Type II.
Type II	-	Solvent shall conform to table I of O-T-620, Type I.
Type III	-	Solvent shall conform to 3.2 of MIL-C-81302, Type II.

3.3.3 Propellant. Propellant shall conform to type 12, table 1 of BB-F-1421. The amount of propellant used shall be adequate to expel all of the solvent from the container.

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3.4 Appearance. The material shall be homogeneous, free of suspended matter, sediment, grit, or other foreign material when tested as specified in 4.4.1.

*3.5 Corrosivity. The material shall not cause any visible pitting, etching, or dark discoloration and no weight change (milligrams/cm^2) greater than 0.5 for magnesium, cadmium and zinc nor greater than 0.2 for aluminum, copper, and brass when tested in accordance with 4.4.7.

*3.6 Effect on plastics, rubber, and paint. The Type I and Type III material shall not show any deleterious effects on plastics, rubber or paint coatings when tested as specified in 4.4.5.

3.7 Functional requirements at low temperature. Electrical contacts, after being treated with Type I, II or III material, shall function properly in environmental temperatures as low as -54°C (-65°F).

*3.8 Leakage. When tested as specified in 4.4.9, the pressurized container shall not leak nor become distorted.

*3.9 Net contents. The average net contents per can shall not be less than the amount specified (5.1.1) or indicated in the contract or purchase order.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

*4.2 Classification of inspections. The inspections are classified as follows:

(a) Materials inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

*4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the component materials used in the cleaning and lubrication compound are in accordance with the applicable referenced specifications or requirements prior to blending the final compound.

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*4.4 Quality Conformance Inspection.

*4.4.1 Sampling for inspection. Unless otherwise specified, sampling shall be performed in accordance with MIL-STD-105. The lot size shall be defined as all containers presented for delivery at one time. The sample shall be one filled container.

*4.4.2 Sampling for tests. The inspection level shall be S-1. Under any circumstance no less than three samples will be tested. Any sample having one or more defects, including less than required fill, shall be cause for rejection of the lot.

*4.4.3 Examination of filled container. The sample unit shall be one filled container ready for packaging. Each sample shall be examined for the following defects:

(a) Dents or distortions
(b) Cap missing
(c) Extension tube missing
(d) Markings--incomplete, incorrect, illegible
(e) Not size specified.

*4.4.4 Examination of preparation for delivery. An examination shall be made to determine that preservation, packaging, packing and marking as required by Section 5 are complied with. Defects shall be scored as specified in TableI. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units. The sample unit shall be one shipping container fully prepared for delivery. The lot shall consist of the number of shipping containers offered for inspection at one time.

TABLE I Examination of Preparation for delivery

Examine	Defect
Markings, exterior and interior	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Material not as specified, component missing, damaged, or otherwise defective, container not as specified.
Workmanship	Inadequate application of components such as incomplete closure of case liners, blocking and cushioning inadequate, improper or missing, count, number of aerosol cans per container not as specified.

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*4.4.5 Effect on plastics, rubber, and painted surfaces. Place a sample of cleaner-lubricant or cleaner on separate specimens of polyester materials, rubber and painted surfaces and allow to remain 24 hours. Examine the surfaces for deposition, cracking, swelling, crazing, softening, and any other deleterious effects.

*4.4.6 Determination of net contents. The sample unit shall be one can. The inspection level shall be S-2. The net weight shall be the difference between the gross weight and the tare weight as obtained from the sample handled, as described in 4.2.3. The average net contents per can shall not be less than the amount specified.

*4.4.7 Corrosivity.

*4.4.7.1 Preparation of specimen. Specimens of the following metals shall be used in this test:

Magnesium, QQ-M-44
Cadmium, QQ-A-671
Zinc, QQ-Z-285
Aluminum, QQ-A-250/4
Copper, QQ-C-576
Brass, QQ-B-626

Suggested specimen size is 3 X 1/2 X 1/16 inches.

*4.4.7.2 Immersion. Three specimens of each of the above metals shall be polished to remove pits, burrs, and irregularities from all faces and edges. The panels shall be finished and cleaned as specified in 4.4.7.2.1 and 4.4.7.2.2. After weighing, the specimens shall be placed in screw cap jars with one specimen in each jar. The specimens shall be placed on one of their narrow ends. The specimens shall be covered with enough compound so that the tops of the specimens are at least 1/4 inch below the surface of the compound. The sealed jar shall then be placed in an oven at 130 $\pm$ 2° F for 7 days. Upon completion of the test, any loose corrosion products shall be removed from the specimens by cleaning as specified in 4.7.4. Reweigh the specimens and calculate the weight loss or gain in milligrams per square centimeter.

*4.4.7.2.1 Preparation of test panels (or disks). Panels (or disks) shall have all sharp edges and burrs removed and shall have all holes chamfered to prevent injury in handling. They shall be surface ground and hand polished with a 240 grit silicon carbide or aluminum oxide cloth or paper to produce a surface finish of 10 to 20 microinches (rms). Iron oxide or so-called "wet or dry" papers or cloths shall not be used.

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*4.4.7.2.2 Cleaning of test panels (or disks). The utensils and cloths used in the cleaning of test panels (or disks) shall be clean and free of contamination. Solvents shall be fresh and renewed frequently. In all stages of treatment the handling of panels (or disks) with the bare hands shall be avoided. The panels (or disks) shall not be permitted to contact contaminated surfaces during the cleaning procedure and shall be handled by tongs and hooks during and after dipping. After-polishing, they shall be cleaned with a surgical gauze swab, in a beaker of hot mineral spirits conforming to Grade I of TT-T-291. Cleaning and scrubbing shall be followed by dipping in (1) a second container of hot mineral spirits, (2) boiling 95 percent methanol, and (3) boiling absolute methanol. The panels (or disks) shall be allowed to dry and shall then be stored in a desiccator until ready for use. If storage of more than 24 hours occurs, the surface preparation shall be repeated starting with the hand polishing.

*4.4.8 Functional characteristics. The supplier shall certify that electrical contacts treated with Type I, II or III material shall remain operative at -54°C (65°F).

*4.4.9 Leakage. Using a filled pressurized can, spray the solvent for three seconds and pause for two seconds. Repeat this operation five times. The can shall then be submerged completely in water maintained at 130 ± 2°F. Let the can remain for five minutes. The emission of bubbles shall indicate leakage. At the end of the five minutes, remove the can from the water and examine it for distortion.

5. PREPARATION FOR DELIVERY

5.1 Dispenser.

*5.1.1 Size. The size of the dispenser shall be sufficient to contain 4, 6, or 16 ounces (avoirdupois), or 113, 169, or 454 grams of the product without exceeding 90 percent at 70°F of the interval volume of the dispenser.

5.1.2 Type. The dispenser shall be of the aerosol type and shall have a finger pressure type dispensing device to produce the desired spray, so designed to shut off when the finger is released. The device shall be protected against activation by use of a snug-fitting metal or plastic protective cap. This cap should provide a clearance of 1/8 inch (32mm) over the dispensing device. A 5-inch (127mm) plastic extension designed to fit into the dispensing device for the purpose of directing the spray into hard to reach places, shall be attached to the outside of the dispenser can by any suitable method, conforming to Class 2, Type IX of PPP-C-96.

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*5.2 Packaging. Packaging shall be Level A.

*5.3 Packing. Packing shall be Level A, or C as specified.

*5.3.1, Level A. Twelve dispensers of contract cleaner, lubricant, or cleaner material shall be packaged in boxes conforming to PPP-B-636, Type I, Class I, Style RSC, individually nested in snug-fitting, full-height, half-slotted style partitions. The partitions shall be made of the same materials as used in the box. Each box shall be water-proof, sealed and closed in accordance with Appendix to PPP-B-636.

*5.3.2 Level C. The contact cleaner-lubricant or cleaner in the dispensers shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with rules and regulations applicable to the mode of transportation.

5.4 Marking.

5.4.1 Civil agencies. In addition to special markings required by the contract or purchase order, the packages and shipping containers shall be marked in accordance with FED-STD-123.

5.4.2 Military agencies. In addition to any special marking required by the contract or purchase order, the packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.4.3 Special marking (precautionary). Each container and each unit package shall bear warning information, marked directly to the surface of the unit containers with marking material which is not affected by the cleaning compound. The warning shall be as follows or reasonable equivalent:

CAUTION: Contents under pressure.

Do not puncture.

Use with adequate ventilation.

Keep away from direct sunlight, radiators, hot water and other heat.

Avoid prolonged or repeated breathing of vapors or contact with skin.

Do not take internally.

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5.4.3.1 Each unit shall be marked as specified by lithograph or other permanent process in accordance with normal commercial practice. The use of gummed or glued labels is prohibited.

6. NOTES

6.1 Intended use. The contact cleaner-lubricant or cleaner material is to be used in cleaning and lubricating electrical contacts. Care should be exercised when using Type II material on synthetic materials as it may affect some of such products under certain conditions. Also Type II material component (methyl chloroform) contains stabilizers and inhibitors which break down under certain conditions giving the possibility of residue deposits remaining after the material has evaporated.

6.1.1 The intended use of this compound does not overlap that of MIL-C-81302. The MIL-C-83360 compound shall be used for the cleaning and lubricating of electrical contacts. The MIL-C-81302 compound shall be used for the cleaning only of precision parts and assemblies and for use as a medium in testing the cleanliness of components that are assumed to be clean.

6.2 Ordering data. Purchasers should exercise any desired options permitted herein and purchase orders should specify the following:

a. Title, number and date of this specification.

b. Level of packaging and packing required.

c. Inspection responsibility if other than as specified herein.

d. Additional marking if required.

6.3 General. It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If for any particular purpose, however, samples with bids are necessary, they should be specifically asked for in the invitation for bids; the specification to apply in all other respects.

6.4 Marginal indication. The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This is done as a convencience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-83360A(AS)	2. DOCUMENT DATE (YYMMDD) 750327

3. DOCUMENT TITLE
CLEANING AND LUBRICATING COMPOUND, ELECTRICAL CONTACT

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial: (2) DSN: (*If Applicable*)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER NAVAL AIR WARFARE CENTER AIRCRAFT DIVISION	b. TELEPHONE NUMBER (*Include Area Code*) (1) Commercial: (908) 323-7488 (2) DSN: 624-7488
c. ADDRESS (*Include Zip Code*) CODE SR3 HIGHWAY 547 LAKEHURST, NJ 08733-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office, 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

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